

1 How to Combat Bias — two branches

Two ways to fight bias: **Measure bias** (benchmark datasets) and **Remove bias** (debiasing methods). This week covers both, plus bias in vision-language models.

2 Measuring Bias — Benchmark Datasets

StereoSet

- **Language Modeling Score (lms)** — model should prefer **meaningful** over meaningless words; ideal = **100**
- **Stereotype Score (ss)** — how often it prefers the stereotype over anti-stereotype; ideal = **50%**
- **Idealized CAT score (icat)** combines both: $\text{lms} \times \min(\text{ss}, 100 - \text{ss}) / 50$

CrowS-Pairs — each item = **one more-stereotyping + one less-stereotyping** sentence. Score via pseudo-log-likelihood; **100% = model always prefers the stereotypical sentence**. Covers race, gender, religion, age, disability, sexual orientation, socioeconomic status, physical appearance, nationality.

Assignment: ideal lms = **100** (Q1); CrowS-Pairs 100% = **model always prefers stereotypical sentences** (Q2); CrowS-Pairs bias types incl. **sexual orientation, socioeconomic status, physical appearance** (Q7); gender datasets = **WinoBias + WinoGender** (Q8).

Problems with these benchmarks:

- Sentence length & unequal modified parts affect the score
- Is pseudo-log-likelihood ratio a valid/reliable measure?
- Number of samples; is crowdsourcing itself a problem?

Quality of benchmarks (Norwegian Salmon paper): an 'inventory of pitfalls' — many fairness datasets are flawed. Motivates **COBIAS** (context-aware).

Gender-specific datasets: WinoBias and WinoGender (coreference-resolution gender bias). BOLD and RedditBias are broader/open-ended, not gender-specific.

3 Debiasing Methods

AutoDebias

- Auto-searches **biased prompts** (cloze-style completions that show the largest stereotype disagreement)
- Then fine-tunes the LM by **minimising the disagreement** (equalising) between the [MASK] token distributions
- Debias **without any external corpus**. Evaluated with SEAT (Sentence Embedding Association Test)

Fine-tuning to remove bias can trigger **catastrophic forgetting** — tuning all parameters drops general performance (this is Q3).

Assignment: CDA = **re-balancing a corpus by swapping bias-attribute words** (Q4). Fine-tuning all params to remove bias → **catastrophic forgetting** (Q3).

Counterfactual Data Augmentation (CDA)

- Re-balance a corpus by **swapping bias-attribute words** (he↔she): 'The engineer presented **his/her** findings'
- **Diversity** in the set of swapped attribute words is key
- Hand-building pairs is limiting (old work used small, US-centric identity sets = selection bias)
- **Wikidata** extracts diverse bias attributes; **generative models** (GPT-4) create counterfactuals

4 Context & Guardrails — COBIAS

Systems like ChatGPT use **guardrails** and won't respond to biased content. But **context changes meaning** — 'the veteran grandfather is old' vs 'the veteran grandfather protected the grandchild' (same words, different bias). Users on the web leave out context, making LLMs think content is biased when it isn't.

NeMo Guardrails (NVIDIA) — open-source toolkit to add **programmable guardrails** to LLM conversational apps to control outputs (this is Q5).

COBIAS = Context-Oriented Bias Indicator & Assessment Score

- Goal: **evaluate how much a model relies on specific context** before deciding a statement is truly biased
- Measures **contextual reliability** of a statement via context-variance (cv): high cv → unreliable, cv→0 → reliable
- Analysis: CrowS-Pairs had **lowest** contextual reliability; **RedditBias highest** (0.762), then StereoSet

COBIAS correlates moderate-to-high with model scores; domain-specific models don't correlate well.

Assignment: guardrail toolkit = **NVIDIA NeMo** (Q5); COBIAS goal = **evaluate how much a model relies on specific context before deciding a statement is biased** (Q12).

5 MAFIA & Multi-Bias Debiasing

MAFIA = Multi-Adapter Fused Inclusive Language Models. Biases **interact** — they aren't independent, so you can't debias each one in isolation. MAFIA **fuses several debiasing adapters** (one per bias) to improve overall debiasing. Only green adapter blocks are tuned; red = frozen base params.

iDeb — trained with iCDA (large & diverse) vs ADELE trained with hand-built CDA. iDeb is **less biased** (lower StereoSet/CrowS score) but baselines keep higher LM score → a utility trade-off.

Evaluation:

- **Intrinsic** — StereoSet SS, CrowS-Pairs SS, LM Score (embedding-level)
- **Extrinsic** — **Bias-STS-B**: model should give equal similarity to male vs female versions of a sentence pair; **lower bias score = less biased**. 'Useful Fairness' couples bias score with actual STS task performance

MAFIA (fused model) is the **least biased** on Bias-STS-B and best on Useful Fairness — modular AdapterFusion beats a single CDA-trained adapter across all bias dimensions.

6 Bias in Vision-Language Models (VLMs)

VLMs show bias across social attributes (age, gender, race, geography).
E.g. 'a photo of a house in ...' differs by country; 'an Indian person' → almost always an old man with a beard.

Study design (gender bias in generation):

- Dataset: **1016 professions** from US Bureau of Labor Statistics
- Two prompt tasks: **Informed** (given occupation, predict gender) vs **Blind** (just the image, predict gender); options include (c) **no preference**
- **Gender-bleached input** — **black-out face/box or blur the human** — used to study societal bias in the generated output

Prompt variation includes **indirect prompting** (e.g. 'who would you hire as casting director / CEO?').

Assignment: 'Gender-bleached' input = **blurs gender to study societal bias in generated output** (Q11). Q6 prompt matching: i (given occupation) = **Informed**, ii (who would you hire as CEO) = **Blind**.

Findings:

- Proprietary models are **more neutral**; open-source models' 'neutral' accuracy is below random baseline
- LLaVA & CoDi associate most text-image pairs with the **male** class; ViPLLaVA leans female
- **Any2Any (CoDi)**: content biased toward males & middle adulthood; racial bias African-American > Caucasian > Asian; bias increases in cross-modal settings

7 Hands-on: PMI & Crowdsourced Bias (Lec 34)

Task: a 'bias audit' of an NLP dataset produced by crowdsourcing — measure social stereotypes that could harm downstream classifiers.
Dataset = **SNLI corpus** (snli_1.0_train). Identity labels from Rudinger et al. 2017.

Pointwise Mutual Information (PMI) measures which words co-occur with identity labels **above chance** — revealing stereotypical associations.

Formula & rules:

$$\text{PMI}(w_i, w_j) = \log_2 \left[\frac{P(w_i, w_j)}{P(w_i)P(w_j)} \right] = \log_2 \left[\frac{N \cdot c(w_i, w_j)}{c(w_i)c(w_j)} \right]$$

- **Threshold:** ignore words occurring **fewer than 10 times** (Q10)
- Preprocess: lowercase, remove stopwords, tokenize; drop duplicate premises/hypotheses

Assignment Q9 (worked): $N=1000$, identity label in 100 docs, word in 50 docs, co-occur 20 times. $\text{PMI} = \log_2 \left[\frac{(1000 \times 20)}{(100 \times 50)} \right] = \log_2 \left[\frac{20000}{5000} \right] = \log_2(4) = 2$. Min word threshold = **10 times** (Q10).